From Measurement Overlap to Shared Influence: Reliable Context Design for MRI Slice Interpolation

Anonymous CVPR submission

Paper ID *****

Abstract

Output disagreement is widely used as a reliability signal, yet two inaccurate reconstructions can agree when they inherit the same error. We study this blindness in deterministic MRI inter-slice interpolation and ask which measurements make neural error observable. For HAT and SwinIR, we define neural shared influence *by applying the same small, seeded perturbation to each physically shared anchor and comparing centered output responses. Across fastMRI brain T2, 115 IXI brain T2 examinations, and patient-clustered NingXia knee T2, neural influence predicts error alignment, shared fraction, and observable fraction more strongly than integer overlap in all six dataset–architecture strata; absolute-correlation advantages are 0.036–0.089 and all 18 clustered intervals exclude zero. Local mechanism arrows remain stable after examination and geometry control, whereas the final context-ranking arrow is heterogeneous. Holding overlap fixed at one confirms continuous influence on fastMRI/IXI but not NingXia. We audit a frozen acquired-anchor cross-validation score against strong adjacent-slice baselines: it ranks x4 error strongly (Spearman 0.943/0.955/0.958) but contains no consistent unique information beyond adjacent difference. It requires zero neural calls and 0.42–3.97 ms CPU time. Frozen IXI+fastMRI x4-to-x2 analyses show positive gains at 10/20/30/50% budgets, without score superiority or prospective-policy claims. The supported principle is narrower and stronger than a count law: reliability depends on shared neural measurement influence, while inexpensive acquired-data triage is useful but not uniquely optimal.*

1. Introduction

Deterministic through-plane interpolation synthesizes missing MRI slices from sparse acquired anchors, but normally provides no indication of when a plausible image is wrong. Ensembles, test-time augmentation, and cross-context consistency convert output variation into reliability evidence [3, 4, 10]. Their blind spot is common error: agreement observes $e_a - e_b$, not either reconstruction error itself.

We investigate how the measurement geometry of MRI interpolation governs this blindness. Our starting observation is that phase-shifted, measurement-disjoint contexts rank error better than contexts that reuse anchors. An exact analytical ladder had already falsified a tempting count-only law: the transition from zero to one shared anchor can be non-monotone. But analytical coefficient weights do not establish what HAT or SwinIR actually use. We therefore define and evaluate a neural finite-difference influence quantity tied to common physical anchors. Across a broad context grid and a cleaner fixed-count subset, it tests whether continuous neural sensitivity explains error alignment beyond integer overlap.

The mechanism motivates a deployment-facing score audit. *Acquired-anchor cross-validation* (Anchor-CV) withholds an interior acquired anchor, reconstructs it from adjacent anchors, and measures its residual. The operation is not new—leave-one-slice-out interpolation predates our work—and we freeze it before Plan6 evaluation. Its value, if any, must survive adjacent-anchor MAE/RMSE, scale, spacing, curvature, spectral, output-PC, and risk–coverage comparisons. We treat falsification of unique-information or superiority claims as part of the contribution.

Our contributions are: (1) a fixed neural shared-influence approximation evaluated on three cohorts, two architectures, and three physical planes per examination; (2) separate tests of influence→alignment, alignment→suppression, and influence→observable fraction, with a heterogeneous final ranking arrow; (3) a fixed-one-overlap test that confirms influence beyond count on fastMRI/IXI and rejects universality on NingXia; and (4) a frozen Anchor-CV explanation, benefit, failure, and efficiency audit showing strong cheap triage but no unique-information or universal-best claim. Dense slices enter only retrospective evaluation.

2. Related Work

Inter-slice MRI reconstruction. SAINT introduced spatially aware medical slice synthesis [15]; I3Net combines inter- and intra-slice information [16]. Single-stack slice-

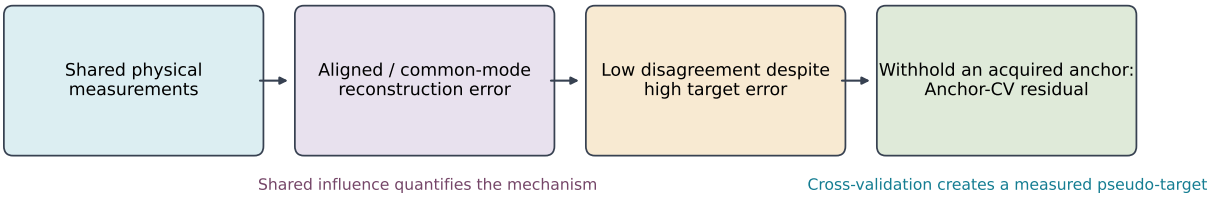


Figure 1. From failure mechanism to deployment-valid score. Reusing influential physical measurements produces error components that two contexts share, so their difference can be small even when both are wrong. Anchor-CV creates a measured pseudo-target from the already acquired lattice instead of relying on another potentially correlated prediction.

to-volume reconstruction jointly treats motion and interpolation [17]. Classical shifted multi-slice super-resolution acquires complementary measurements [5], while acquisition design has compared shifted and rotated schemes under Bayesian accuracy and precision [13]. These works improve reconstruction; we design reliability contexts for a frozen interpolator.

MRI uncertainty and active acquisition. Learned MRI uncertainty includes variational/Bayesian reconstruction and deep ensembles [9, 12]. Active acquisition selects additional k-space measurements to reduce learned uncertainty [18]. Our setting is through-plane image-space interpolation. We neither learn a posterior nor optimize k-space: we study when output differences can observe error and derive a score from already acquired slice anchors. Hu *et al*. showed that uncertainty variation can miss systematic MRI reconstruction error [7]; we therefore do not claim consensus blindness itself as novel. Likewise, leave-one-slice-out evaluation was used for medical interpolation [6]. Our novelty audit retains the physical measurement-dependence intervention, local neural shared-influence approximation, and frozen deployment-oriented evaluation role—not the algebra or holdout operation.

Consistency and shared error. Predictive diversity interacts with individual error rather than replacing it [2, 14]. MRI self-consistency, such as SPIRiT, enforces compatibility among acquired measurements [11]. Our distinction is intervention-level: the physical measurements supporting the same missing target are explicitly controlled, and their estimator influence is related to target-error alignment.

3. Disagreement Observability

For target slice y and deterministic predictions $f_a = y + e_a$, $f_b = y + e_b$,

$$\|f_a - f_b\|^2 = \|e_a\|^2 + \|e_b\|^2 - 2\langle e_a, e_b\rangle. \tag{1}$$

Writing $\bar{e} = (e_a + e_b)/2$ and $\delta = (e_a - e_b)/2$ gives the exact decomposition

$$\tfrac{1}{2}(\|e_a\|^2 + \|e_b\|^2) = \|\bar{e}\|^2 + \|\delta\|^2. \tag{2}$$

We call the normalized terms the shared and observable fractions. If $e_a = e_b \neq 0$, disagreement is zero despite error; if $e_a = -e_b$, all paired error energy is observable. Thus diversity is not calibration, and low disagreement cannot certify correctness.

3.1. Shared Measurement Influence

Let a local estimator be linear in its physical measurements, $f = \sum_j c_j x_j$. Two contexts are embedded in the union of physical anchor positions, yielding coefficient vectors c_a, c_b with zeros at unused positions. Raw overlap counts common nonzero positions but ignores their influence. We test three acquisition-only summaries: coefficient cosine, signed dot product, and common absolute mass

$$I_\cap(a, b) = \sum_j \min(|c_{a,j}|, |c_{b,j}|). \tag{3}$$

All are target-free geometry/estimator properties for a linear estimator. They motivate but do not define influence for a nonlinear neural network. Let $F_c(x)$ denote a frozen interpolator under context c. For physical anchor j shared by contexts a, b, apply the same seeded centered Rademacher perturbation $\epsilon\xi_{js}$ at 1% of the globally normalized case range. The centered finite-difference response is

$$r_{cjs} = F_c(x + \epsilon\xi_{js}) - F_c(x). \tag{4}$$

We average the response cosine $\langle r_{ajs}, r_{bjs}\rangle/(\|r_{ajs}\|\|r_{bjs}\|)$ over shared anchors and four frozen seeds (6101–6104). This *neural shared influence* is a local sensitivity approximation around the observed input. It is not a global attribution theorem and does not identify influence from unshared anchors.

This definition separates three concepts that an integer count conflates. Two contexts may reuse one weak or one

influential anchor; both receive the same overlap count but yield different shared responses. Response cosine measures directional alignment, while common response mass retains magnitude. Agreement among alternatives is a robustness check. Finite differences were chosen because they directly perturb acquired intensities and can be applied identically to HAT and SwinIR; no heatmap-based qualitative selection was used.

4. Methods

4.1. Cohorts and Reconstruction Outputs

The manifest contains 24 held-out fastMRI brain T2 cases [8], 22 out-of-fold NingXia knee T2 series from eight patients, and 115 held-out IXI brain T2 cases [1]. Frozen HAT and SwinIR x4 predictions are evaluated; fastMRI/NingXia also have x2 outputs for retrospective densification benefit. NingXia inference resamples patient prefixes with all subordinate series retained. Physical planes are matched with NIfTI affines. Dense targets synthesize retrospective anchors and evaluate errors but never enter a reliability score. Evaluation uses a fixed foreground mask. All scores are reduced to one value per examination before cohort-level ranking, preventing long volumes from dominating the endpoint.

4.2. Clean Overlap Controls

The neural zero-overlap condition reconstructs the same missing plane from phase-0 and phase-2 x4 lattices. The clean maximal-overlap context applies horizontal flip equivariance to identical anchors, preserving z order, distance, count, mask, and preprocessing. The primary endpoint is Spearman correlation between masked prediction disagreement and paired MAE, with 5,000 examination/patient-cluster bootstrap replicates.

For an exact ladder, four anchors have target-relative absolute distances $\{1, 2, 3, 4\}$. A reference sign pattern $(+, -, +, -)$ is compared with every sign pattern sharing exactly $k \in \{0, 1, 2, 3, 4\}$ positions and retaining anchors on both sides. A distance-weighted local linear regression, weights $1/d$, predicts the same target. We average all valid sign configurations at each k; no favorable arrangement is selected. A wider $\{2, 4, 6, 8\}$ ladder is a depth-limited replication.

For the neural experiment, one central eligible target defines a reference context with offsets $(-2, +2)$ and eight target-preserving comparators. Target, input count, model, mask, and preprocessing are fixed; overlap, span, asymmetry, and physical identity are recorded. A registered stage gate expanded to three planes per examination only after every cohort/architecture had the expected endpoint direction and at least two cohorts improved over count. Confirmation therefore covers 72 fastMRI, 345 IXI, and 66 NingXia model-specific planes per architecture. The grid is broad rather than perfectly distance matched, so partial rank analyses control span, asymmetry, and examination.

A cleaner subset retains the four comparator contexts with exactly one shared anchor. Integer overlap is then constant. Neural influence is tested after controlling span, asymmetry, and examination. This is the strongest test that influence contains information beyond count; failure in NingXia is retained rather than averaged away.

4.3. Acquired-Anchor Cross-Validation

Given acquired x4 anchors $x_0, x_4, x_8, \ldots$, each interior x_{4j} is withheld and predicted linearly from $x_{4(j-1)}$ and $x_{4(j+1)}$. Anchor-CV is the median masked MAE between predicted and actually acquired withheld anchors. It is model-independent and available before any denser acquisition. We compare case ranking with adjacent-anchor RMSE, analytical x8 bipartition disagreement, and the extra-grid phase score. Full allocation curves use 10/20/30/50% budgets and include PC, TTA, committee, maximal-shared, random expectation, and oracle where compatible.

The median aggregation is prespecified to tolerate isolated motion or boundary failures. Importantly, Anchor-CV does not feed the withheld residual into the reconstruction model and does not assert exchangeability of individual planes. It asks a narrower question: whether an examination's acquired through-plane signal is locally predictable at the native sparse spacing. Retrospective correlation with x4 error and x4-to-x2 improvement tests whether that examination-level property is useful for triage.

Plan6 freezes the original identity-foreground masked half-second-difference implementation as ANCHOR_CV_V1. We reproduced every stored score from the frozen formula (maximum absolute difference 5.68×10^{-14} after matching float32 masks) before examining new outcomes. Strong baselines include adjacent-anchor MAE/RMSE/SSIM, gradient and frequency differences, robust intensity range, spacing, curvature, first-difference variability, and output PC. Partial rank association conditions on adjacent MAE, scale, and spacing; leave-one-cluster-out ridge tests whether feature stacking explains the dominant examination effect without in-sample tuning.

IXI $\times 2$ evidence was generated only after freezing checkpoint, SHA-256, preprocessing, metric, and 10/20/30/50% budgets. The 115-case run is definition matched but cross-estimator: benefit is mean HAT/SwinIR $\times 4$ error minus I3Net $\times 2$ error. It tests case-level retrospective consequence, not a same-model causal treatment effect.

4.4. Statistics

The independent units are 24 fastMRI examinations, 115 IXI examinations, and eight NingXia patients. Plane results are descriptive; uncertainty uses 5,000 clustered bootstrap replicates. Paired bootstrap intervals compare Anchor-CV Spear-

man to locked comparators. No cohort-specific score transform or favorable budget selection is used. All new experiments are exploratory but were registered in the local experiment registry before their outputs were computed. The full mechanism chain is evaluated arrow by arrow. Influence-to-error-alignment and influence-to-observable-fraction partial ranks control examination, span, and asymmetry. Alignment-to-normalized-disagreement additionally conditions on influence. Context-level ranking correlates each context's mean alignment/influence with its disagreement–error Spearman; with few contexts this final arrow has lower power. We describe the chain as conditional association, never causal mediation. The supplement reports every architecture stratum, frozen protocol, risk curve, budget interval, and material negative result.

5. Results

5.1. Clean Sharing Produces Blindness

Zero overlap ranks error better than the clean maximal-overlap control in all cohorts: the Spearman differences are 0.092 (95% CI 0.043–0.176) on fastMRI, 0.193 (0.111–0.371) on NingXia, and 0.095 (0.078–0.126) on IXI. Clean-max error cosine is 0.949, 0.995, and 0.968, leaving observable error fractions of only 2.61%, 0.25%, and 1.58%. Under zero-derived fixed quartile thresholds, clean sharing increases high-error/low-disagreement prevalence by 27.1%, 28.2%, and 20.9%, with every clustered interval above zero.

The original order-reversal control supports the complete covariance chain on NingXia and IXI: overlap raises covariance and shared fraction while lowering observable fraction, disagreement, and ranking. Several arrows reverse on fastMRI, confirming that reversal is confounded and motivating the clean control.

5.2. An Exact Ladder Falsifies the Count-Only Law

Figure 2 shows all compact ladder levels. Overall trends have the predicted signs in every cohort: overlap-count trend is $+0.9$ for error cosine/shared fraction, -0.9 for observable fraction, and -0.9 to -1.0 for disagreement. However, every cohort violates strict monotonicity at $0 \to 1$: one shared anchor slightly lowers error cosine and increases observable fraction. Reliability correlation remains 0.958–0.992 for every nonidentical level. The wide ladder replicates the exception in IXI while NingXia is strictly monotone. Therefore count is associated with the mechanism but is not its sufficient causal dose.

5.3. Neural Influence Explains More Than Count

The registered stage gate passed, so the neural experiment expanded to three planes per examination. Figure 3 and Table 2 summarize the confirmatory endpoints. Neural influence had stronger absolute association than overlap count

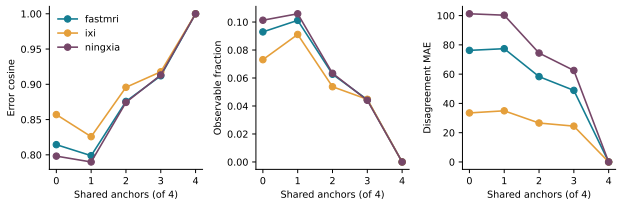


Figure 2. Exact compact four-anchor ladder. Target, measurement count, absolute distances, estimator, and mask are fixed. Overall trends are strong, but the preserved $0 \to 1$ exception falsifies strict count monotonicity.

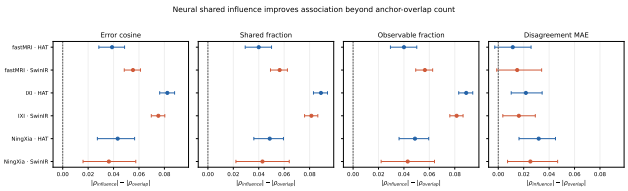


Figure 3. Neural finite-difference shared influence versus integer overlap. Forest intervals show the absolute-correlation advantage for error alignment, shared fraction, observable fraction, and supporting disagreement across frozen architectures/cohorts.

with error cosine, shared fraction, and observable fraction in all six dataset–architecture strata. The advantage was 0.036–0.089; all 18 examination/patient-cluster bootstrap intervals excluded zero. Rank-residual correlation of neural influence after overlap, span, asymmetry, and examination control remained $|\rho| = 0.185$–0.395.

The result is not driven by one network. HAT/SwinIR influence-to-error-cosine correlations were 0.838/0.851 on fastMRI, 0.797/0.787 on IXI, and 0.786/0.772 on NingXia. Corresponding overlap-count correlations were 0.799/0.796, 0.714/0.712, and 0.743/0.735. Disagreement magnitude improved only modestly over count, and both fastMRI advantage intervals crossed zero. We therefore treat direct disagreement as supporting, not confirmatory, evidence.

5.4. Fixed-Count Test Rejects Universality

Holding overlap exactly at one shared anchor removes count as an explanation. On fastMRI and IXI, neural influence retained the expected partial association with every endpoint/model after span, asymmetry, and examination control: $|\rho| = 0.135$–0.587 and all clustered intervals excluded zero. For example, error-cosine partial ρ was 0.169/0.350 on fastMRI HAT/SwinIR and 0.346/0.258 on IXI.

NingXia did not replicate the fixed-count result. Error-cosine partial ρ was 0.014 for HAT and -0.039 for SwinIR; every endpoint interval crossed zero and several directions reversed. The broad-grid advantage in NingXia therefore does not establish continuous influence beyond count/geometry under eight-patient fixed-count inference. We reject a uni-

Table 1. Clean same-target overlap control. $\Delta\rho$ is zero-overlap minus clean-maximal-overlap error-ranking correlation. Observable fraction is the paired error energy exposed by prediction difference. Stable-wrong is the prevalence increase under zero-derived fixed quartile thresholds. Intervals are clustered at examination/patient level.

Cohort	Independent units	$\Delta\rho$ [95% CI]	Clean-max error cosine	Clean-max observable / stable-wrong Δ
fastMRI brain T2	24 examinations	.092 [.043, .176]	.949	2.61% / +27.1 pp
IXI brain T2	115 examinations	.095 [.078, .126]	.968	1.58% / +20.9 pp
NingXia knee T2	8 patients (22 series)	.193 [.111, .371]	.995	0.25% / +28.2 pp

Table 2. Influence versus overlap for error cosine

Cohort	Model	ρ_{count}	ρ_{neural}	Adv.
fastMRI	HAT	.799	.838	.039
	SwinIR	.796	.851	.055
IXI	HAT	.714	.797	.082
	SwinIR	.712	.787	.075
NingXia	HAT	.743	.786	.043
	SwinIR	.735	.772	.036

versal law and retain two supported claims: neural influence improves the broad descriptor in all cohorts, and beyond-count evidence is clean only in the two larger cohorts.

5.5. The Local Chain Is Stronger Than the Ranking Chain

The full mechanism was evaluated as four separate arrows. Influence associated with error alignment after examination and geometry control in all six strata ($\rho = 0.219$–0.410); alignment associated with suppressed normalized disagreement conditional on influence/geometry ($\rho = -0.933$ to -0.889); and influence associated with lower observable fraction ($\rho = -0.412$ to -0.211). All 18 local-arrow intervals excluded zero.

The final context-level consequence was weaker. Greater alignment predicted poorer disagreement–error ranking in all strata, but intervals excluded zero only for both IXI models and NingXia HAT. Influence-to-ranking was less stable still. This asymmetry prevents a causal-mediation claim: finite-difference sensitivity explains local error geometry, while a small set of contexts and case mixtures determine whether that geometry changes a global ranking statistic.

The first and fourth arrows should not be conflated. A local context pair can have aligned errors because common influential anchors drive both outputs. Whether a *set* of contexts then yields worse global case ranking depends on the distribution of error magnitude, target anatomy, and the finite number of context configurations. Strong local cancellation is necessary evidence for the proposed mechanism but is not sufficient for a universal context-level ranking theorem.

5.6. Anchor-CV Is Strong but Not Unique

The frozen implementation correlates with x4 reconstruction error by 0.943, 0.955, and 0.958 on fastMRI, IXI, and NingXia (Fig. 4); Kendall correlations are 0.812/0.831/0.844. However, adjacent MAE/RMSE are statistically indistinguishable and adjacent MAE is slightly higher on IXI/NingXia. Anchor-CV partial association given adjacent MAE, robust scale, and spacing is $0.333/-0.059/-0.100$. Scale-normalizing the score sharply reduces or reverses ranking. ANCHOR_CV_V1 is an interpretable acquired-data difficulty score, but the claim that leave-one-anchor reconstruction adds unavailable information is falsified.

The explanation audit distinguishes five hypotheses. Intrinsic through-plane difficulty (H1/H2) is supported by the cross-architecture case effect and leave-one-cluster-out prediction. Invisible consensus error (H3) is only indirectly supported: Anchor-CV tracks total error but is not an error-map estimator. Raw scale (H4) is a major contributor because normalization degrades the score. Proxying spacing/direct change (H5) is also supported because partial ranks vanish on IXI/NingXia. No single universal curvature/spectral feature transfers, and combining correlated features overfits IXI calibration.

An examination score is still useful without unique information if it is cheap, frozen, and transparent. But method selection changes: adjacent MAE is the default zero-call baseline, Anchor-CV is an alternate acquired residual with the same order of cost, and output PC is appropriate only when multiple reconstructions already exist. The paper therefore evaluates capability under resource constraints rather than promoting a new state-of-the-art scalar.

Under the frozen IXI+fastMRI consequence test, Anchor-CV predicts retrospective x4-to-x2 improvement with $\rho = 0.948$ and 0.871, respectively. Adjacent MAE gives 0.950/0.846 and output PC 0.970/0.843. All policies have positive clustered gain above random at 10/20/30/50% budgets. IXI Anchor-CV captures 0.920/0.936/0.972/0.973 of oracle gain (Fig. 5); fastMRI captures 0.998/1.000/0.876/0.933. No score dominates. Within-case localization remains unsupported: Anchor-CV is examination triage, not a slice heatmap.

Absolute IXI Anchor-CV benefit exceeds random by 26.53, 23.12, 21.73, and 14.41 MAE units at 10/20/30/50%

Table 3. Fixed-one-overlap neural influence partial Spearman after span, asymmetry, and examination control. Parentheses are clustered 95% intervals for error cosine; O is observable fraction and D is disagreement.

Cohort	Model	Error cosine	Shared fraction	Observable O	Disagreement D
fastMRI	HAT	.169 [.053,.283]	.135	−.135	−.161
	SwinIR	.350 [.212,.464]	.321	−.321	−.286
IXI	HAT	.346 [.259,.424]	.360	−.360	−.543
	SwinIR	.258 [.181,.335]	.241	−.241	−.587
NingXia	HAT	.014 [−.232,.223]	−.006	.006	−.102
	SwinIR	−.039 [−.175,.144]	−.039	.039	−.072

Table 4. Mechanism-chain partial Spearman ranges across HAT/SwinIR. Every local-arrow interval excludes zero. Context-ranking support counts architecture strata whose interval excludes zero.

Cohort	Influence→alignment	Alignment→norm. disagreement	Influence→observable	Alignment→ranking
fastMRI	.372–.410	−.933–−.926	−.412–−.370	−.333–−.143 (0/2 supported)
IXI	.321–.347	−.893–−.890	−.342–−.298	−.643–−.500 (2/2 supported)
NingXia	.219–.258	−.898–−.889	−.231–−.211	−.667–−.548 (1/2 supported)

Table 5. Frozen x4 error ranking audit. Partial ρ conditions on adjacent MAE, robust scale, and spacing.

Cohort	Anchor ρ	Anchor τ	Adjacent ρ	Partial ρ
fastMRI	.943	.812	.934	.333
IXI	.955	.831	.956	−.059
NingXia	.958	.844	.973	−.100

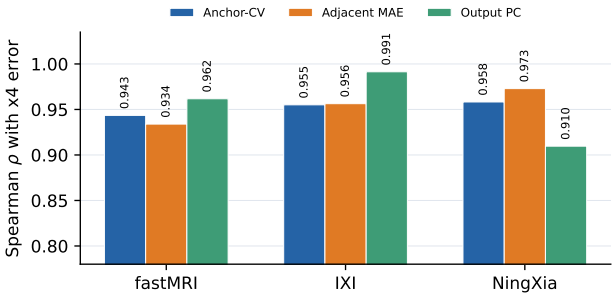


Figure 4. Frozen examination-level x4 error ranking. Anchor-CV is strong, but adjacent MAE and an output principal component are statistically competitive.

budgets; all clustered lower limits are positive. fastMRI corresponding gains are 14.73, 12.49, 8.30, and 5.50. These are averages over selected cases, not population treatment effects. The union of measurements needed for some output scores may itself approach denser coverage, while Anchor-CV and adjacent difference require only acquired anchors. Budget curves therefore measure retrospective ranking under a fixed outcome table, not scanner-time utility.

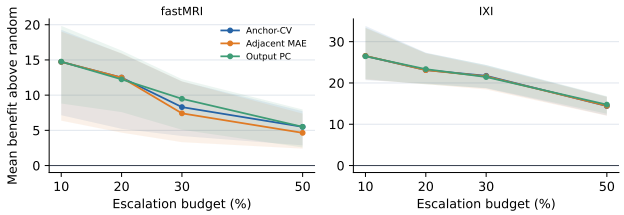


Figure 5. All prespecified IXI+fastMRI budgets. Lines show gain above random selection and bands are patient-cluster bootstrap 95% intervals. Gains are retrospective.

5.7. Case Difficulty, Risk Curves, and Calibration

HAT–SwinIR case-error ranks are 0.989/1.000/0.997 on fastMRI/IXI/NingXia. In a two-architecture squared-error decomposition, examination identity accounts for 99.1%/99.99%/99.78%, while architecture after case accounts for 0.19%/0.004%/0.029%. Thus architecture-independent difficulty is not an interpretation added after Anchor-CV succeeds; it is the dominant empirical variation.

Leave-one-examination/patient-cluster-out ridge using Anchor-CV alone achieves rank correlation 0.939/0.954/0.953 and mean-baseline R^2 0.789/0.900/0.837. Adding direct difference, curvature, spacing, and spectral features does not improve ranks and produces negative IXI calibration R^2. Observable acquired quantities explain much of case ordering, but naive correlated feature stacking is rejected.

Complete frozen x4 risk curves reinforce non-superiority (Fig. 6). Output PC has lower nAURC than Anchor-CV on fastMRI (HAT/SwinIR .015/.014 versus .020/.031) and IXI (.0012/.0012 versus .0149/.0148). Adjacent MAE is strongest on NingXia (.0057/.0100 versus Anchor

Table 6. Frozen x4→x2 benefit allocation. Values are selected mean benefit minus random expectation; all clustered 95% intervals exclude zero. Oracle fraction is shown for Anchor-CV.

Cohort	Score	10%	20%	30%	50%	Anchor oracle fraction range
fastMRI	Anchor-CV	14.73	12.49	8.30	5.50	.876–1.000
	Adjacent MAE	14.73	12.49	7.41	4.64	–
	Output PC	14.76	12.24	9.48	5.50	–
IXI	Anchor-CV	26.53	23.12	21.73	14.41	.920–.973
	Adjacent MAE	26.56	23.12	21.60	14.47	–
	Output PC	26.47	23.35	21.44	14.75	–

Table 7. Frozen Anchor-CV fraction of oracle gain above random.

Budget	10%	20%	30%	50%
fastMRI	.998	1.000	.876	.933
IXI	.920	.936	.972	.973

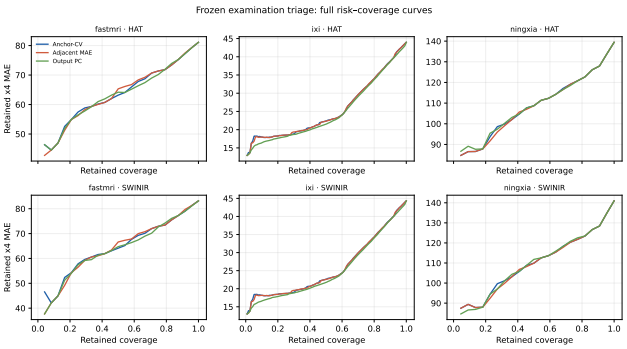


Figure 6. Frozen x4 risk–coverage. Anchor-CV is a strong acquired-only score, while output PC or adjacent MAE yields lower nAURC depending on cohort.

.0094/.0138). Anchor-CV remains useful at every cohort, but no evidence supports calling it the best risk score.

5.8. Failure Taxonomy and Efficiency

With a predeclared 0.20 absolute rank-error margin, PC is closer to true x4 error in three fastMRI and eight IXI model–cases; Anchor-CV is closer in one fastMRI model–case; NingXia has no large-margin discordance. Most observations are concordant (44/48, 222/230, 44/44). The rare IXI PC winners have lower adjacent difference and dynamic range than cohort medians, but sample counts do not justify a new fusion mechanism. The supplementary MRI/reconstruction/error panels show that neither examination score localizes the worst plane.

Efficiency explains why a non-superior score may still be operationally distinct. Anchor-CV CPU time is 0.42/3.97/3.93 ms on representative fastMRI/IXI/NingXia examinations, with zero neural calls and zero predicted-volume storage. Committee, flip-TTA, and disjoint pairs each require two calls; sequential totals are 0.25–0.29 s, 2.43–3.20 s, and 0.49–0.60 s by cohort, about 271 MiB peak allocated GPU memory, and 8.6–66.5 MiB output storage. Output PC score-only latency is small only after two predictions exist. Cost is confirmed; accuracy superiority is not.

Table 8. Representative scoring strategy cost

Strategy	Calls	Time range	Storage
Anchor-CV	0	0.42–3.97 ms	0
Committee	2	0.25–2.43 s	10–67 MiB
Flip-TTA	2	0.26–3.18 s	10–67 MiB
Disjoint pair	2	0.29–3.20 s	9–66 MiB

6. Discussion

The experiments refine a coarse observation into a neural mechanism and a scoped capability. Measurement sharing aligns reconstruction errors and hides them from disagreement, but simply counting shared anchors is inadequate. Identically perturbing common physical anchors reveals how similarly each frozen network responds. This local neural influence improves explanation of error geometry over count in every broad-grid stratum and survives an exact fixed-count test in the two larger cohorts.

Why this is more than niche MRI. The measured anchors are imaging-specific, but the distinction between shared information and shared output influence applies to any paired reliability estimator whose members reuse observations. Two predictors can have the same overlap count yet different local sensitivities, and a perturbation that changes appearance may preserve the relevant error direction. MRI makes the dependence physically auditable because anchor identity and target position are known. Generalization to other inverse problems is a hypothesis, not an evaluated claim.

Why finite differences. Input gradients, integrated gradients, occlusion, and leave-one-anchor-out could all define sensitivity. Finite differences were selected before endpoint computation because they test the actual frozen input–output map, permit identical perturbations in two contexts, and avoid assuming differentiability through preprocessing. Centered seeded perturbations reduce a single-direction artifact. The remaining dependence on amplitude and random field is a limitation; a future study should test scale linearity and structured physical perturbations.

Anchor-CV does not complete a clean "mechanism produces a superior method" story. It is cheap and strongly ranks examination difficulty and densification consequence, but adjacent difference explains nearly all of its rank information and output PC wins IXI benefit ranking. This negative audit is useful: the method contribution is computationally distinct, not statistically unique. Its failure to localize planes further limits it to examination triage.

The work does not estimate calibrated uncertainty, select a prospective scanner sequence, or establish clinical safety. Finite differences are local to the observed input, perturbation distribution, and amplitude; they do not identify a global neural Jacobian or causal mediation. Comparator contexts vary span/asymmetry, and the fixed-count NingXia result is null/mixed under only eight patients. IXI x2 is cross-estimator. A prospective policy should freeze the pilot, cost model, and threshold, then evaluate new acquisitions without dense-target leakage.

What the negative results change. A strict "less overlap is always better" prescription is unsupported by the analytical $0 \to 1$ exception. A universal "neural influence beyond count" law is unsupported by fixed-count NingXia. An "Anchor-CV is uniquely informative/best" claim is falsified by adjacent MAE and output PC. The remaining contribution is sharper: continuous neural influence is a better broad context descriptor, locally supports the alignment/cancellation mechanism, and acquired-only triage offers a low-cost option without dominance.

Deployment interpretation. Anchor-CV can be computed immediately after a sparse examination has been reconstructed and before deciding whether additional through-plane sampling is worth its cost. The present evidence supports ranking a batch of examinations under a fixed budget, not an autonomous reacquisition decision. Scanner time, motion risk, anatomy, and clinical urgency are absent from the utility function. A prospective trial would need to combine those costs with a frozen score and evaluate realized, rather than retrospective, benefit.

Patient-level inference. NingXia contributes 22 series but only eight independent patients. Every clustered interval carries all series, models, planes, and contexts of a patient together. This produces wider and sometimes null fixed-count results despite hundreds of output comparisons. Treating series or planes as patients would have made the mechanism appear artificially universal. We retain NingXia as an informative heterogeneous boundary, not an equally powered confirmation cohort.

Efficiency and failure. Anchor-CV takes 0.42/3.97/3.93 ms CPU on fastMRI/IXI/NingXia, with zero neural calls and no predicted-volume storage. Committee, flip-TTA, and disjoint pairs require two calls, 0.25–0.29/2.43–3.20/0.49–0.60 s, about 271 MiB peak GPU allocation, and 8.6–66.5 MiB outputs. Output PC is cheap only after two predictions exist. Failure taxonomy finds few large disagreements between PC and Anchor-CV: PC is more accurate for 3 fastMRI and 8 IXI model–cases, Anchor-CV for one fastMRI, and neither has large-margin NingXia wins. Most cases are concordant; no fusion is justified.

Reproducibility and scope. Each claim is linked to a registered experiment, manifest, per-case/configuration output, and clustered summary. The finite-difference seeds, amplitude, contexts, checkpoint identities, and IXI x2 hash were frozen. We report the failed fixed-count NingXia result, scale dependence, feature-stacking failure, rare score discordance, and failed slice localization. No local association is presented as a global theorem.

7. Conclusion

Disagreement reveals only error components that differ between reconstruction contexts. In MRI inter-slice interpolation, local neural shared influence predicts error alignment and observable fraction more precisely than shared-anchor count across three broad context grids, with beyond-count confirmation limited to fastMRI and IXI. Frozen acquired-anchor cross-validation cheaply ranks error and retrospective benefit but is not uniquely informative or universally best. Reliability design should reduce shared influence, test the complete local chain, and preserve negative cohort boundaries; agreement alone cannot certify correctness.

References

[1] Biomedical Image Analysis Group, Imperial College London. IXI dataset, 2014. Accessed 2026-08-13. 3

[2] Gavin Brown, Jeremy Wyatt, Rachel Harris, and Xin Yao. Diversity creation methods: A survey and categorisation. *Information Fusion*, 6(1):5–20, 2005. 2

[3] Vineet Edupuganti, Morteza Mardani, Shreyas Vasanawala, and John M. Pauly. Uncertainty quantification in deep mri reconstruction. *IEEE TMI*, 40(1):239–250, 2021. 1

[4] Yarin Gal and Zoubin Ghahramani. Dropout as a bayesian approximation: Representing model uncertainty in deep learning. In *ICML*, pages 1050–1059, 2016. 1

[5] Hayit Greenspan, Gozde Oz, Nahum Kiryati, and Shlomo Peled. Mri inter-slice reconstruction using super-resolution. *Magnetic Resonance Imaging*, 20(5):437–446, 2002. 2

[6] Antal Horvath, Simon Pezold, Matthias Weigel, Katrin Parmar, and Philippe C. Cattin. High order slice interpolation for medical images. In *SASHIMI*, pages 69–78, 2017. 2

[7] Shi Hu, Nicola Pezzotti, and Max Welling. Learning to predict error for mri reconstruction. In *MICCAI*, pages 604–613, 2021. 2

[8] Florian Knoll, Jure Zbontar, Anuroop Sriram, et al. fastMRI: A publicly available raw k-space and dicom dataset of knee images for accelerated mr image reconstruction using machine learning. *Radiology: Artificial Intelligence*, 2(1): e190007, 2020. 3

[9] Thomas Kuestner, Kerstin Hammernik, Daniel Rueckert, Tobias Hepp, and Sergios Gatidis. Predictive uncertainty in deep learning-based mr image reconstruction using deep ensembles. *Magnetic Resonance in Medicine*, 92(1):289–302, 2024. 2

[10] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *NeurIPS*, 2017. 1

[11] Michael Lustig and John M. Pauly. SPIRiT: Iterative self-consistent parallel imaging reconstruction from arbitrary k-space. *Magnetic Resonance in Medicine*, 64(2):457–471, 2010. 2

[12] Dominik Narnhofer, Alexander Effland, Erich Kobler, Kerstin Hammernik, Florian Knoll, and Thomas Pock. Bayesian uncertainty estimation of learned variational mri reconstruction. *IEEE TMI*, 41(2):279–291, 2022. 2

[13] Michele Nicastro, Ben Jeurissen, Quinten Beirinckx, Celine Smekens, Dirk H. J. Poot, Jan Sijbers, and Arnold J. den Dekker. To shift or to rotate? comparison of acquisition strategies for multi-slice super-resolution magnetic resonance imaging. *Frontiers in Neuroscience*, 16:1044510, 2022. 2

[14] Luis A. Ortega, Rafael Cabanas, and Andres Masegosa. Diversity and generalization in neural network ensembles. In *AISTATS*, pages 11720–11743, 2022. 2

[15] Cheng Peng, Wei-An Lin, Haofu Liao, Rama Chellappa, and S. Kevin Zhou. SAINT: Spatially aware interpolation network for medical slice synthesis. In *CVPR*, pages 7750–7759, 2020. 1

[16] Haofei Song, Xintian Mao, Jing Yu, Qingli Li, and Yan Wang. I3net: Inter-intra-slice interpolation network for medical slice synthesis. *IEEE TMI*, 43(9):3306–3318, 2024. 1

[17] Sean I. Young, Yael Balbastre, Bruce Fischl, Polina Golland, and Juan Eugenio Iglesias. Fully convolutional slice-to-volume reconstruction for single-stack mri. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11535–11545, 2024. 2

[18] Zizhao Zhang, Adriana Romero, Matthew J. Muckley, Pascal Vincent, Lin Yang, and Michal Drozdzal. Reducing uncertainty in undersampled mri reconstruction with active acquisition. In *CVPR*, pages 2049–2058, 2019. 2